T H A I M I N H N H A T

E M B E D D E D E N G I N E E R

Tan Binh Ward, Ho Chi Minh City | 0387201803 | thaiminhnhat1832003@gmail.com | www.linkedin.com/in/nhatwork1803

PROFILE SUMMARY

Driven and technically adept final-year Mechatronics Engineering student with 1.5 years of hands-on R&D experience in embedded systems. Experienced in developing robust firmware, communication protocols, and integrated hardware-software solutions. Proven track record of owning projects from system architecture to deployment, supported by strong analytical thinking and a continuous learning mindset. TOEIC 975.

EXPERIENCE

❖ **247 EQUIPMENT CENTER TRADING & SERVICE CO. LTD.** **SEP 2024 - NOW**

Embedded Engineer.

Smart Shelf Scale System.

- Designed the system architecture for a multi-node setup with 32 input sensors, using multiplexers to expand load cells capacity.
- Designed a custom wireless communication protocol using nRF24L01 RF with error checking using CRC7 polynomial and encryption using base64 and handshake.
- Achieved target measurement accuracy across different load ranges with ±1g @ 0~2 kg, ±5g @ 2~5 kg, ±10g @ 5~10 kg.
- Developed core control functions and data processing logic for the system.
- Built an automated calibration pipeline for individual sensors.
- Designed 2-layer PCBs for both slave and master nodes using Altium Design and Kicad.
- Built a desktop application to manage and operate the system.
- Authored full documentation: user manual, technical reference, internal docs.

Tech stack: Arduino Mega 2560, HX711 load cell, nRF24L01, 74HC4067 Multiplexer, UART, custom RF protocol, Altium, KiCad, Python/Tkinter.

❖ **Z ROBOTICS CO. LTD.** **JUN 2024 - JUL 2024**

Research and Development Intern.

- Configured Marlin firmware for a custom 5-axis 3D printer using MKS Robin Nano v3.1.
- Modeling printing process with CoppeliaSim.

Tech stack: C, Marlin firmware, CoppeliaSim.

❖ **ALTA TECHNOLOGY AND COMMUNICATION SERVICES LTD.** **JUN 2023 - JUL 2023**

Robot Research and Development Intern.

- Programming Dobot CR5 robot arm.
- Simulation robot's kinematic using Dobot SC Studio.

Tech stack: C/ C++, Dobot SC Studio.

PROJECT

❖ **IMPLEMENTING REINFORCEMENT Q-LEARNING IN TRAFFIC SIGNAL OPTIMIZATION FOR ONE INTERSECTION.** **MAR 2024 - DEC 2025**

- Design a tabular Q-Learning model to optimize traffic signal control of a junction.
- Simulate and evaluate model and envinronment using SUMO - gui (Simulation of Urban MObility) and Python

Tech stack: - Tabular Q-learning, SUMO - gui, Python.

❖ **SHORT-TERM TRAFFIC FLOW FORECASTING WITH XGBOOST AND TRAFFIC CONDITION CLASSIFICATION VIA GRADIENT BOOSTING MACHINES.** **MAR 2024 - APR 2024**

- Using XGBoost and Gradient Boosting Machines algorithm to predict the traffic flow.
- Using SUMO - gui (Simulation of Urban MObility) to simulate vehicles of a junction and evaluate parameters of it.

Tech stack: - SUMO - gui, Python.

❖ **DESIGN LINE FOLLOWER ROBOT.** **APR 2024 - JUN 2024**

Description: - Design a line follower robot using PIC16F877.

Responsibilities: - Programming PIC16F877 reading sensors and control motor through PWM.

Technologies used: - PIC16F877, I2C protocol, photoelectric sensor, encoder, PWM.

SKILLS

Language:	English.
Programming:	C/ C++, Python, Embedded C, Bash/ Shell.
Compilers:	Matlab, VS Code Studio, Arduino IDE.
Embedded:	Micro-controller: Arduino, PIC16F, FreeRTOS basics
Protocol:	CAN, I2C, SPI, UART.
Machine learning:	Q-Learning, XGBoost, Gradient Boosting Machines
Tool:	Git (version control basics), Jira, Confluence
PCB design:	Altium, KiCad
Mechatronics:	PLC programing, Matlab Simulink.

EDUCATION

❖ **HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY.** **AUG 2021 - PRESENT**

Major: Mechatronics Engineering

GPA: 3.0/ 4.0

HONORS & AWARDS

❖ **SCHOOL SEMESTER SCHOLARSHIP.** **DEC 2023**

- Scholarship for students with excellent study result during the semester.
- Scoring 3.7/ 4.0

❖ **CK82 SCHOLARSHIP** **DEC 2023**

Scholarship for mechatronic students with high grade.